

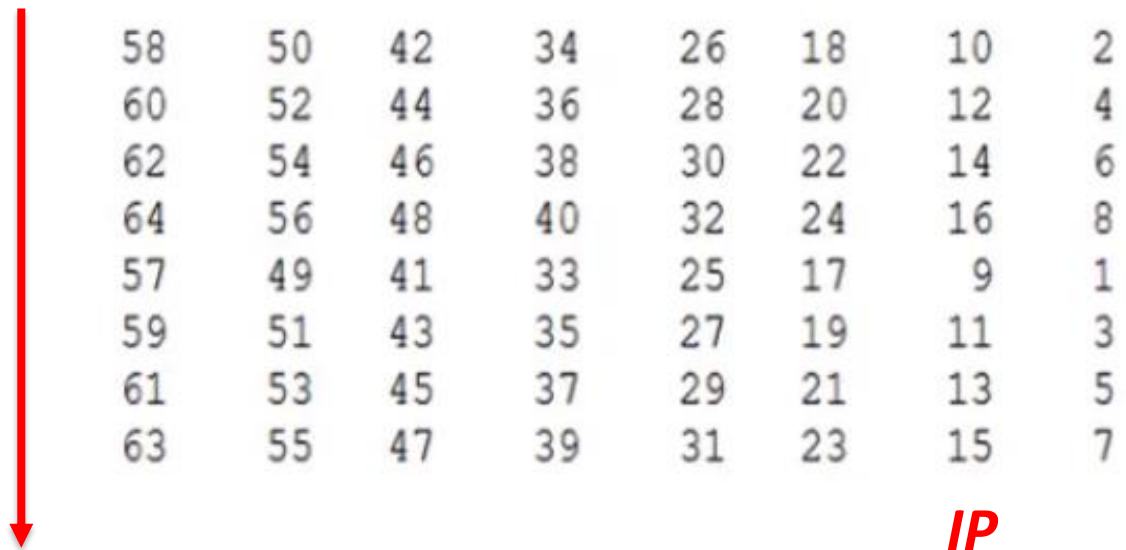
DES Block Cipher Part II

Module II

Step 2: Encoding the plain text 64-bit

M = 0000 0001 0010 0011 0100 0101 0110 0111 1000 1001 1010 1011 1100 1101 1110 1111

There is an Initial permutation (*IP*) of the 64 bits of the message data M.



A diagram illustrating the Initial Permutation (IP) of a 64-bit message. A vertical red arrow on the left points downwards, indicating the mapping from the original bit positions to the permuted bit positions. The mapping is shown as a table with 8 rows and 8 columns. The first column contains the original bit positions (58, 60, 62, 64, 57, 59, 61, 63) and the subsequent columns contain the new bit positions (50, 42, 34, 26, 18, 10, 2, 4, 54, 46, 38, 30, 22, 14, 6, 8, 49, 41, 33, 25, 17, 9, 1, 51, 43, 35, 27, 19, 11, 3, 53, 45, 37, 29, 21, 13, 5, 55, 47, 39, 31, 23, 15, 7). The label *IP* is written in red at the bottom right of the table.

58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

IP

IP = 1100 1100 0000 0000 1100 1100 1111 1111 1111 0000 1010 1010 1111 0000 1010 1010

Step 2: Encoding the plain text 64-bit

IP = 1100 1100 0000 0000 1100 1100 1111 1111 1111 0000 1010 1010 1111 0000 1010 1010

Next divide the permuted block IP into a left half L_0 of 32 bits, and a right half R_0 of 32 bits.



$L_0 = 1100\ 1100\ 0000\ 0000\ 1100\ 1100\ 1111\ 1111$ (32 bits)

$R_0 = 1111\ 0000\ 1010\ 1010\ 1111\ 0000\ 1010\ 1010$ (32 bits)

16 Rounds – first round

$L_0 = 1100\ 1100\ 0000\ 0000\ 1100\ 1100\ 1111\ 1111$ (32 bits)

$R_0 = 1111\ 0000\ 1010\ 1010\ 1111\ 0000\ 1010\ 1010$ (32 bits)

$$L_n = R_{n-1}$$

$$R_n = L_{n-1} + f(R_{n-1}, K_n)$$

Let $+$ denote XOR addition



$n=1$ for round 1

$$L_1 = R_{1-1}$$

$$R_1 = L_{1-1} + f(R_{1-1}, K_1)$$

$$L_1 = R_0$$

$$R_1 = L_0 + f(R_0, K_1)$$

For $n=1$, we have

$$K_1 = 000110\ 110000\ 001011\ 101111\ 111111\ 000111\ 000001\ 110010$$

$$L_1 = R_0 = 1111\ 0000\ 1010\ 1010\ 1111\ 0000\ 1010\ 1010$$

$$R_1 = L_0 + f(R_0 + K_1)$$

$$f(R_0 + K_1)$$

$R_0 = 1111\ 0000\ 1010\ 1010\ 1111\ 0000\ 1010\ 1010$ (32 bits)

To calculate f , we first expand each block R_0 from 32 bits to 48 bits. This is done by using a selection table that repeats some of the bits in R_0 .

If we refer the use of this selection table the function E .

Thus, $E(R_0)$ has a 32 bit input block, and a 48 bit output block.

$E(R_0) = 011110\ 100001\ 010101\ 011110\ 100001\ 010101\ 010101$

E bit selection table

32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1

$$f(R_0 + K_1)$$

$$E(R_0) = 011110 \ 100001 \ 010101 \ 011110 \ 100001 \ 010101 \ 010101$$

Next, calculate f , we have to XOR the output $E(R_0)$ with the key K_1 .
 $K_1 + E(R_0)$

$$K_1 = 000110 \ 110000 \ 001011 \ 101111 \ 111111 \ 000111 \ 000001 \ 110010$$

$$E(R_0) = 011110 \ 100001 \ 010101 \ 011110 \ 100001 \ 010101 \ 010101$$

$$K_1 + E(R_0) = 011000 \ 010001 \ 011110 \ 111010 \ 100001 \ 100110 \ 010100 \ 101111$$

The result is 48 bits or we can say eight group containing six bits.

$$f(R_0 + K_1)$$

We now do the strange with each group of six bits: we use them as *Addresses in tables called “S boxes”*.

$$\begin{array}{cccccccc}
 K_1 + E(R_0) = & 011000 & 010001 & 011110 & 111010 & 100001 & 100110 & 010100 & 101111 \\
 K_1 + E(R_0) = & B1 & B2 & B3 & B4 & B5 & B6 & B7 & B8
 \end{array}$$

We now calculate,

$$S_1(B_1) S_2(B_2) S_3(B_3) S_4(B_4) S_5(B_5) S_6(B_6) S_7(B_7) S_8(B_8)$$

$$f(R_0 + K_1)$$

$$K_1 + E(R_0) = 011000 \ 010001 \ 011110 \ 111010 \ 100001 \ 100110 \ 010100 \ 10111$$

$$K_1 + E(R_0) = \quad B1 \quad \quad B2 \quad \quad B3 \quad \quad B4 \quad \quad B5 \quad \quad B6 \quad \quad B7 \quad \quad B8$$

We now calculate,

$$S_1(B_1) \ S_2(B_2) \ S_3(B_3) \ S_4(B_4) \ S_5(B_5) \ S_6(B_6) \ S_7(B_7) \ S_8(B_8)$$

$$B_1 = 011000$$

$S_1(B_1)$ = row no. = first bit & last bit from B_1 = 00 in binary $\Rightarrow$ 0 in decimal
 column no. = 1100 in binary $\Rightarrow$ 12 in decimal = (0, 12) = 5 = 0101 in binary

$$S_1(B_1) = 0101.$$

S1

s - box

Column Number

Row
No.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
1	0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
2	4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
3	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

$$f(R_0 + K_1)$$

$$S_1(B_1) = 0101.$$

$S_1(B_1)$	$S_2(B_2)$	$S_3(B_3)$	$S_4(B_4)$	$S_5(B_5)$	$S_6(B_6)$	$S_7(B_7)$	$S_8(B_8)$
0101	1100	1000	0010	1011	0101	1001	0111

$f(R_0 + K_1)$ – final stage of f – permutation

$S_1(B_1)$ $S_2(B_2)$ $S_3(B_3)$ $S_4(B_4)$ $S_5(B_5)$ $S_6(B_6)$ $S_7(B_7)$ $S_8(B_8)$
 0101 1100 1000 0010 1011 0101 1001 0111

P

16	7	20	21
29	12	28	17
1	15	23	26
5	18	31	10
2	8	24	14
32	27	3	9
19	13	30	6
22	11	4	25

$f(R_0, K_1) = 0010\ 0011\ 0100\ 1010\ 1010\ 1001\ 1011\ 1011$ (32 bits)

$$f(R_0, K_1) = 0010 \ 0011 \ 0100 \ 1010 \ 1010 \ 1001 \ 1011 \ 1011$$

$$L_0 = 1100 \ 1100 \ 0000 \ 0000 \ 1100 \ 1100 \ 1111 \ 1111$$

$$R_0 = 1111 \ 0000 \ 1010 \ 1010 \ 1111 \ 0000 \ 1010 \ 1010$$

For $n=1$, we have

$$L_1 = R_0 = 1111 \ 0000 \ 1010 \ 1010 \ 1111 \ 0000 \ 1010 \ 1010$$

$$R_1 = L_0 + f(R_0, K_1)$$

$$= 1100 \ 1100 \ 0000 \ 0000 \ 1100 \ 1100 \ 1111 \ 1111 \\ 0010 \ 0011 \ 0100 \ 1010 \ 1010 \ 1001 \ 1011 \ 1011$$

$$R_1 = 1110 \ 1111 \ 0100 \ 1010 \ 0110 \ 0101 \ 0100 \ 0100$$

$$L_1 = 1111 \ 0000 \ 1010 \ 1010 \ 1111 \ 0000 \ 1010 \ 1010$$

After 16 rounds ..

Output of 16 Rounds

$L_{16} = 0100\ 0011\ 0100\ 0010\ 0011\ 0010\ 0011\ 0100$

$R_{16} = 0000\ 1010\ 0100\ 1100\ 1101\ 1001\ 1001\ 0101$

We reverse the order of these two blocks and apply the final permutation to

$R_{16}L_{16} = 00001010\ 01001100\ 11011001\ 10010101\ 01000011\ 01000010\ 00110010\ 00110100$

$IP^{-1} = 10000101\ 11101000\ 00010011\ 01010100\ 00001111\ 00001010\ 10110100\ 00000101$

IP^{-1}

Cipher text:

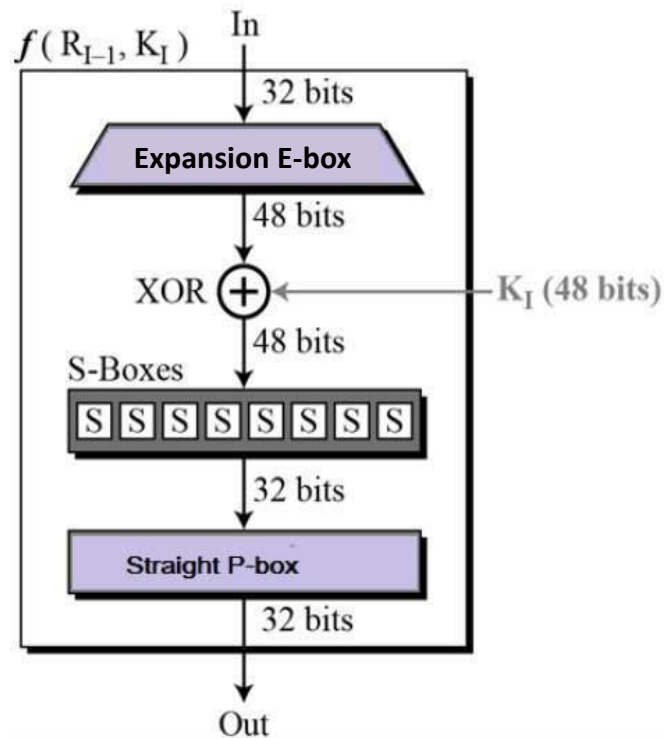
85E813540F0AB405

40	8	48	16	56	24	64	32
39	7	47	15	55	23	63	31
38	6	46	14	54	22	62	30
37	5	45	13	53	21	61	29
36	4	44	12	52	20	60	28
35	3	43	11	51	19	59	27
34	2	42	10	50	18	58	26
33	1	41	9	49	17	57	25

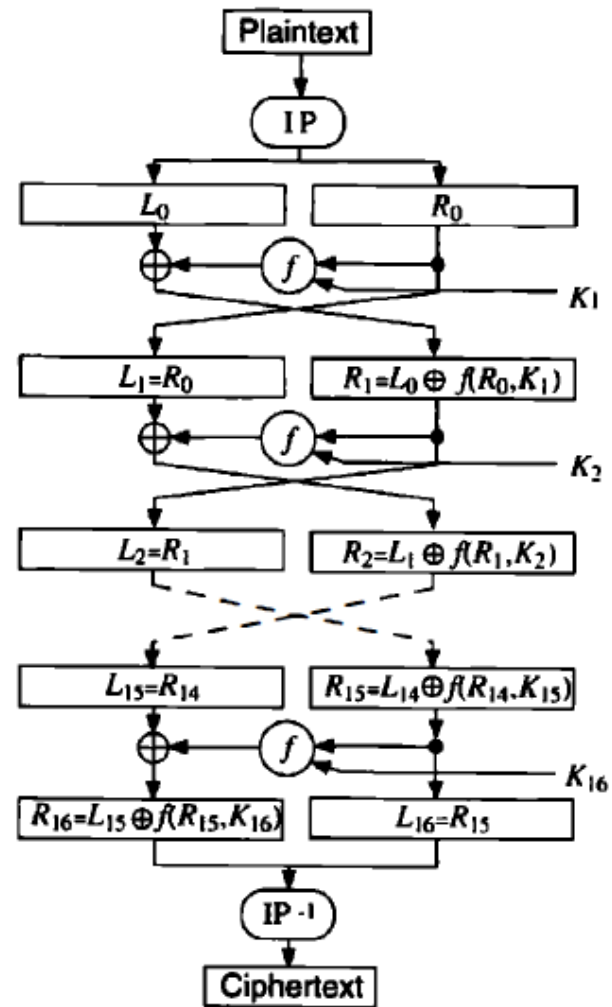
Like that we encrypt all the remaining blocks.

About DES

- DES is insecure due to the relatively short 56-bit key size.
- In January 1999, distributed.net and the Electronic Frontier Foundation collaborated to
- publicly break a DES key in 22 hours and 15 minutes.
- This cipher has been superseded by the Advanced Encryption Standard (AES).
- DES has been withdrawn as a standard by the National Institute of Standards and Technology.



Single Round function (F) of the DES



DES Encryption Flowchart

(a) Initial Permutation (IP)

58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

(c) Expansion Permutation (E)

32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1

(b) Inverse Initial Permutation (IP⁻¹)

40	8	48	16	56	24	64	32
39	7	47	15	55	23	63	31
38	6	46	14	54	22	62	30
37	5	45	13	53	21	61	29
36	4	44	12	52	20	60	28
35	3	43	11	51	19	59	27
34	2	42	10	50	18	58	26
33	1	41	9	49	17	57	25

(d) Permutation Function (P)

16	7	20	21	29	12	28	17
1	15	23	26	5	18	31	10
2	8	24	14	32	27	3	9
19	13	30	6	22	11	4	25

Tables used in the DES algorithm

DES Block Cipher

Key schedule (generator):

This algorithm generates the subkeys ($K \rightarrow K_1, K_2 \dots K_{16}$).

1- The **56 bits** of the key are selected from the initial **64** by Permuted Choice 1 (**PC1**) table.

2- The **56 bits** are divided into **two 28-bit** halves.

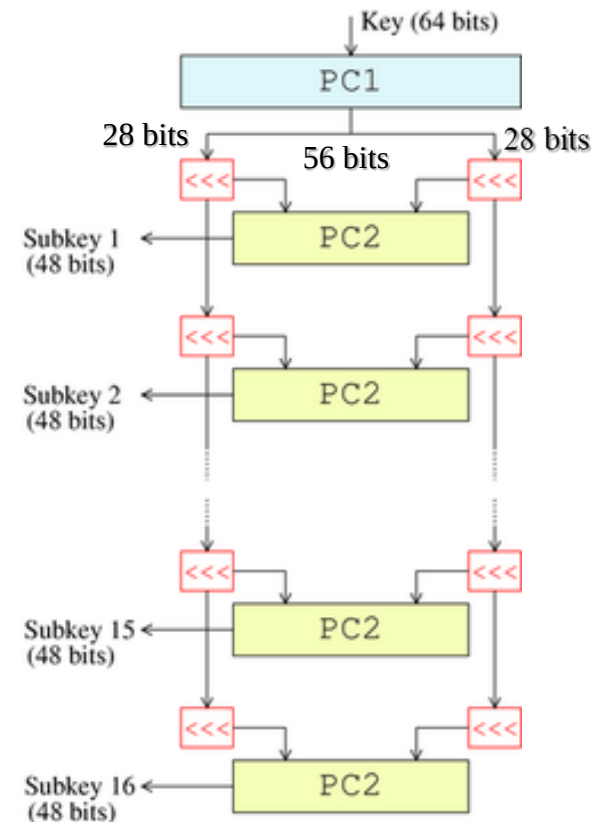
3- In each round, both halves are rotated left by one or two bits (specified for each round).

4- The **48** subkey bits are selected by Permuted Choice 2 (**PC2**) table (**24 bits from the left half, and 24 from the right**) and used in each round.

General remarks in the DES:

1- The S-boxes provide the core of the security of DES and the cipher would be linear, and trivially breakable without them.

2- The substitution and permutation in the DES provide confusion and diffusion.



Key schedule structure

S₁

14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

S₂

15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9

S₃

10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12

S₄

7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15
13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9
10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4
3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14

S₅

2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3

S₆

12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11
10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8
9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6
4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13

S₇

4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1
13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6
1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2
6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12

S₈

13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7
1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2
7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8
2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11

(a) Input Key

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64

(b) Permuted Choice One (PC-1)

57	49	41	33	25	17	9
1	58	50	42	34	26	18
10	2	59	51	43	35	27
19	11	3	60	52	44	36
63	55	47	39	31	23	15
7	62	54	46	38	30	22
14	6	61	53	45	37	29
21	13	5	28	20	12	4

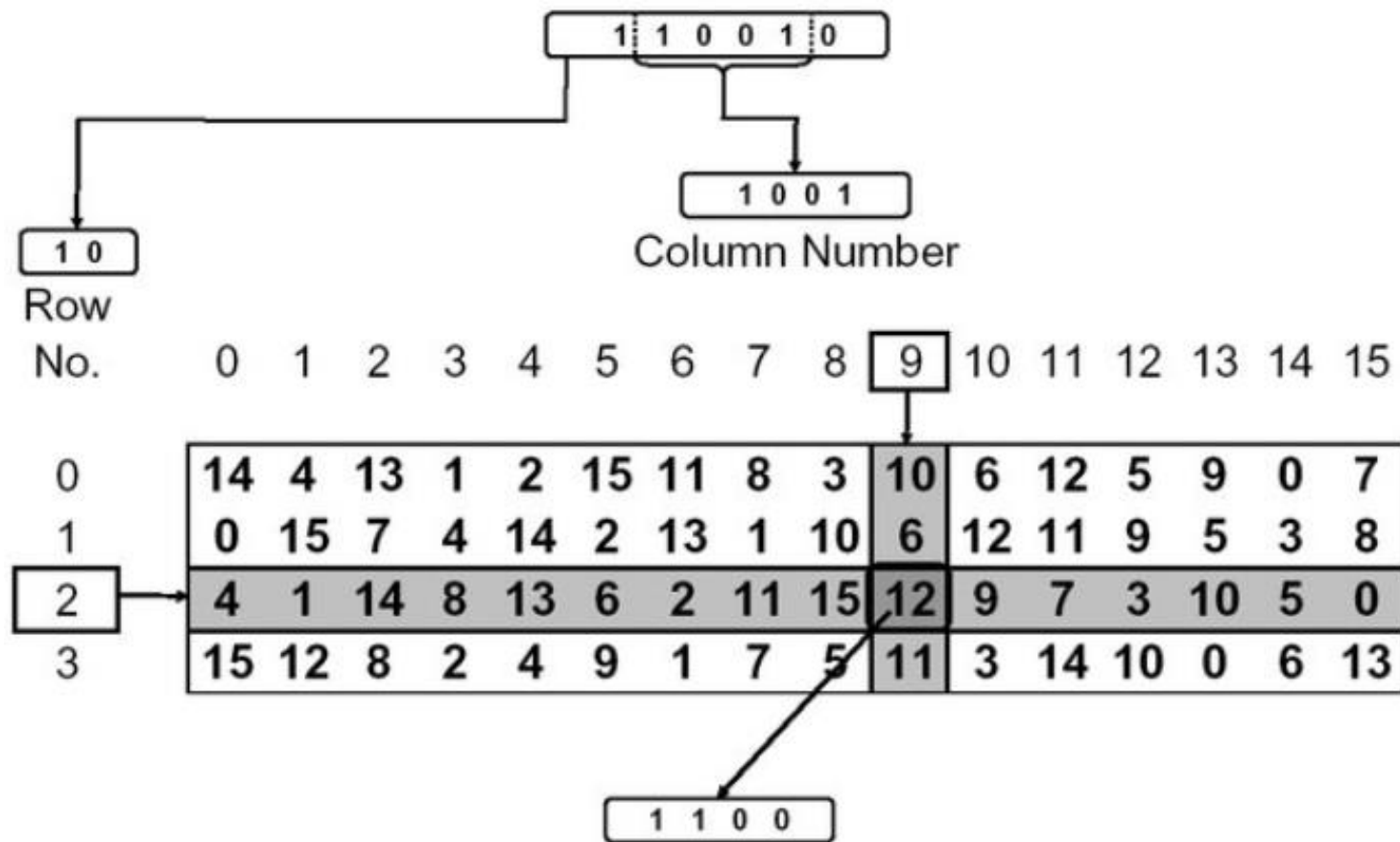
(c) Permuted Choice Two (PC-2)

14	17	11	24	1	5	3	28
15	6	21	10	23	19	12	4
26	8	16	7	27	20	13	2
41	52	31	37	47	55	30	40
51	45	33	48	44	49	39	56
34	53	46	42	50	36	29	32

(d) Schedule of Left Shifts

Round Number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Bits Rotated	1	1	2	2	2	2	2	2	1	2	2	2	2	2	2	1

Tables used in DES key generator



Application of S-box in DES Algorithm

DES Block Decryption

DES decryption :

The decryption algorithm uses the same steps exactly as in the encryption algorithm except that the application of the subkeys is reversed (i.e. in round1 use K_{16} , round2 use K_{15} and so on).

Security and cryptanalysis:

The two most widely used attacks on block ciphers are linear and differential cryptanalysis. DES is also vulnerable to a brute-force (exhaustive search) attack.

Triple DES

In cryptography, Triple DES (3DES or TDES), officially the Triple Data Encryption Algorithm (TDEA or Triple DEA), is a symmetric-key block cipher, which applies the DES cipher algorithm three times to each data block.

Therefore, Triple DES uses a "key bundle" that comprises three DES keys, K1, K2 and K3 , each of 56 bits.

The encryption algorithm is:

$$\text{ciphertext} = E_{k3}(D_{k2}(E_{k1}(\textit{plaintext})))$$

Decryption is the reverse:

$$\textit{plaintext} = D_{k1}(E_{k2}(D_{k3}(\textit{ciphertext})))$$

Each triple encryption encrypts one block of 64 bits of data.